

Lidar E1 - E5

Concept

- Predict **box location** with “Object Detection” then measure the box dimension with lidar. To scope down location of box and cut out lidar noise
- The prediction model trained in-house by Dobybot to be able to detect box with “Black & White” image (Infrared)
- Return of prediction is in (x1,y1,x2,y2) format. Float32.

Full Infrared Image Resolution

640 pixel



480 pixel

Prediction from Machine Learning

Detections

(xyxy=array([[385.79095 , 286.39618 , 460.3058 , 370.37234],
[79.322464, 192.35791 , 254.25697 , 366.5183]]),

“bbox” location.

2 lists means model detect 2 boxes in this image

dtype=float32),

mask=None,

confidence=array([0.5392826, 0.5156908], dtype=float32),

class_id=(1, 1)),

tracker_id=None, data={}, metadata={})

['class_id :1 conf :0.54', 'class_id :1 conf :0.52']

Confidence of detection 1.0 means 100% confidence sequence by “A”

Class of detection.

When training object detection. We can train many objects in the same time. Ex.

Id_1 = Bicycle

Id_2 = Person

Id_2 = car

The model will be able to detect 3 things But downside is model will be larger and slower.

Dobybot train this model specially for **box detection only**.

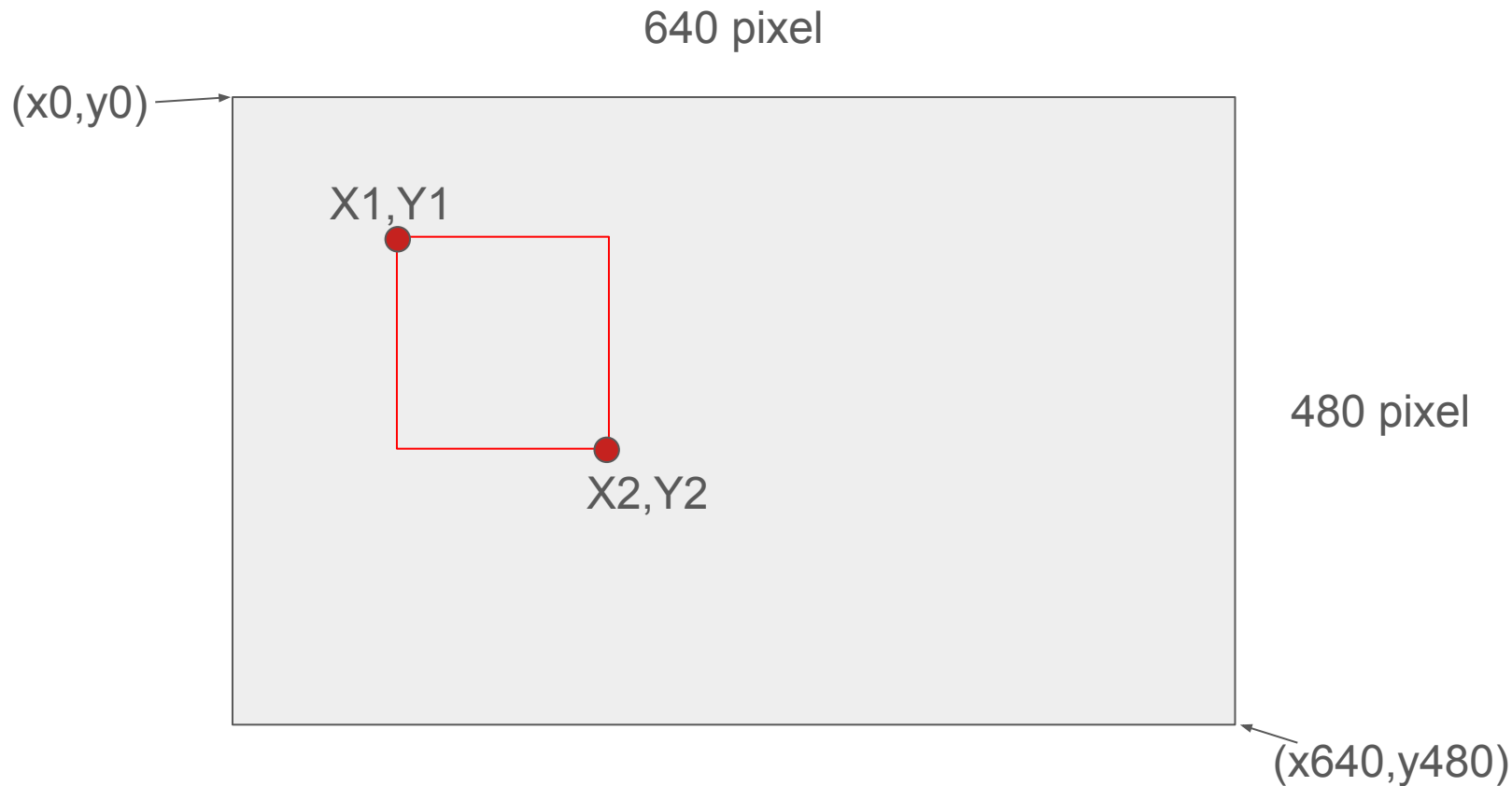
So Id_1 = “Box”

Model is small and faster prediction.

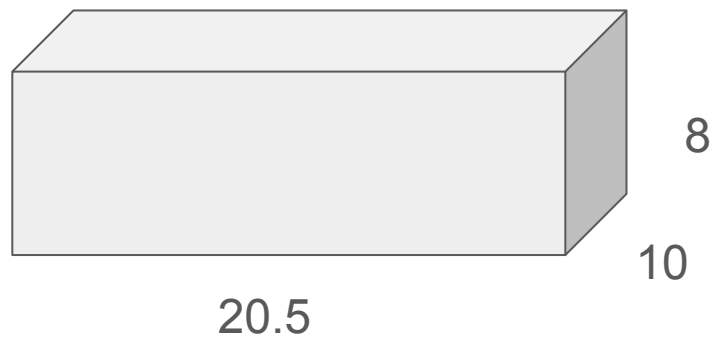
Sequence by “A”

Conclusion of detection

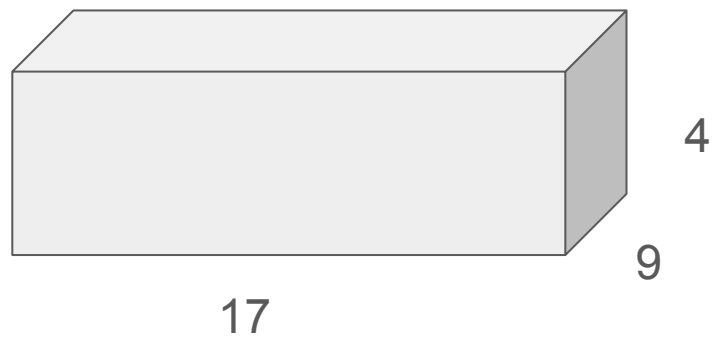
How to read bbox



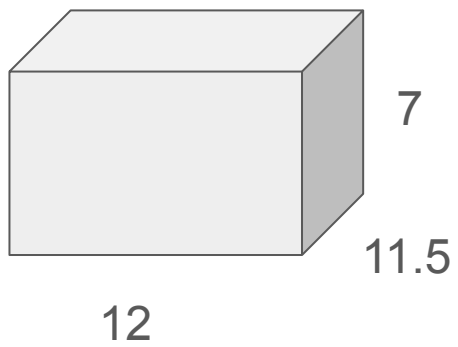
E1



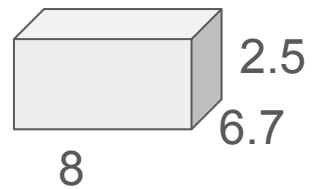
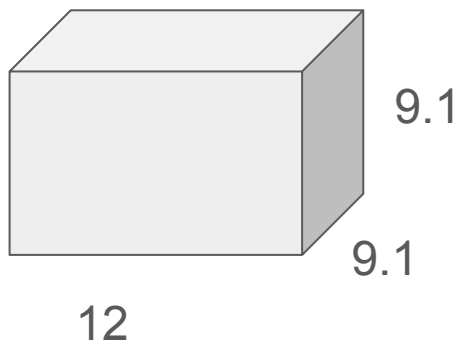
E2



E3



E4



E5

